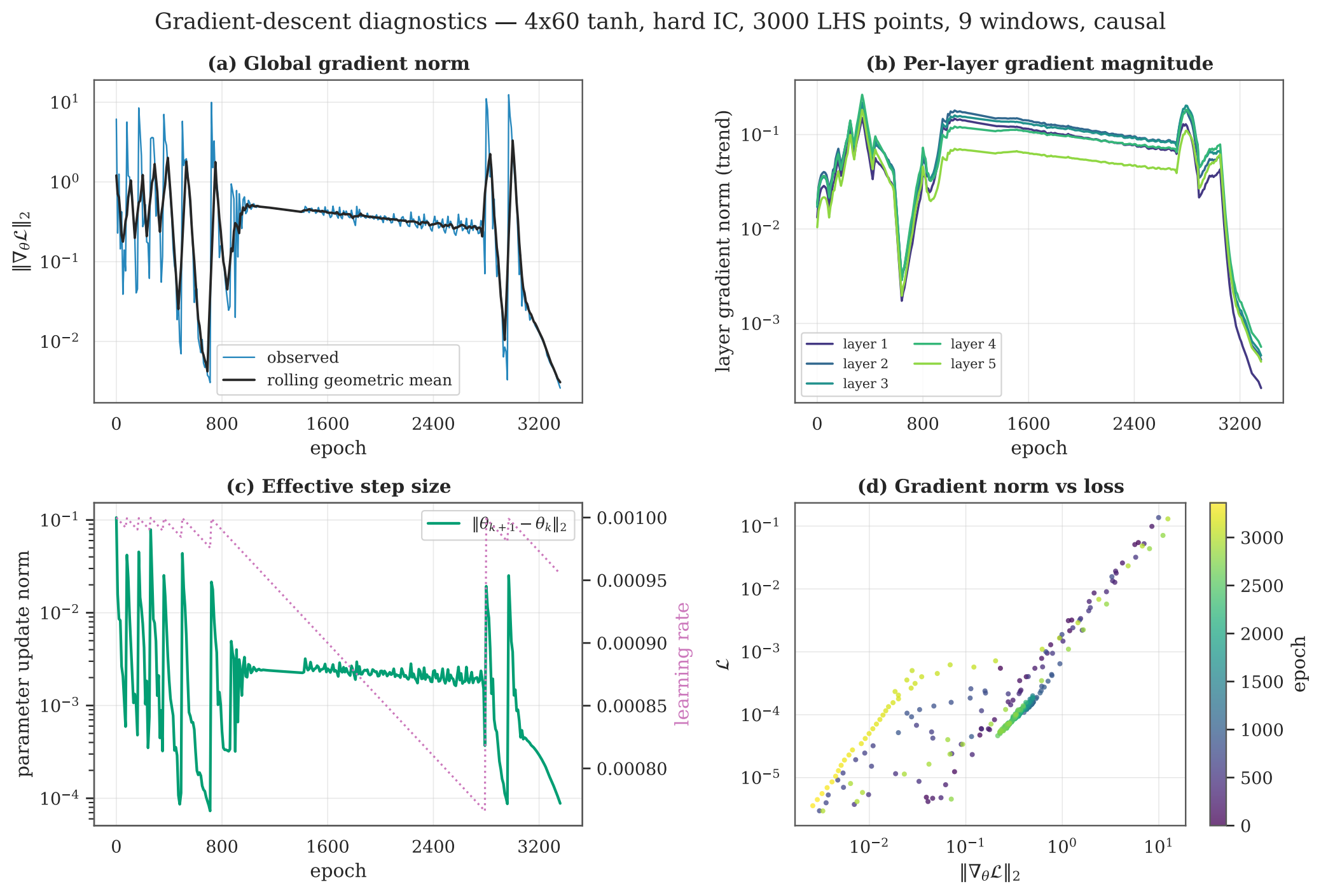

Gradient-descent diagnostics — 4x60 tanh, hard IC, 3000 LHS points, 9 windows, causal
(a) Global gradient norm
$\|\nabla_\theta \mathcal{L}\|_2$
observed
rolling geometric mean
epoch
(b) Per-layer gradient magnitude
layer gradient norm (trend)
layer 1
layer 2
layer 3
layer 4
layer 5
epoch
(c) Effective step size
parameter update norm
$\|\theta_{k+1} - \theta_k\|_2$
learning rate
epoch
(d) Gradient norm vs loss
$\mathcal{L}$
$\|\nabla_\theta \mathcal{L}\|_2$
epoch